

Analytical Chemistry (CHEM-319)

2020

(ii)

Ans The fraction of solute is extracted is equal to the millimoles of the solute in the organic layer divided by the total number of millimoles of solute. The millimoles are given by the molarity times the milliliters. Thus, the percent extracted is given by:

$$\%E = \frac{[S]_o [V]_o}{[S]_o [V]_o + [S]_a [V]_a} \times 100\%$$

where V_o and V_a are the volumes of the organic and solution phase respectively, and S_o and S_a is the concentration of the distributing species of organic and aqueous layer respectively.



(v)

Ans The Nernst distribution law, also known as the Nernst Partition law or Nernst equation, describes the distribution of a solute between two immiscible solvents at equilibrium.

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It states that the ratio of the concentrations of a solute in two immiscible solvents is constant at a given temperature. The law is expressed as:

$$[A]_1 / [A]_2 = K_d$$

Where:

$[A]_1$ is the concentration of solute in first solvent.

$[A]_2$ is the concentration of solute in second solvent.

K_d is the distribution coefficient or partition coefficient, which is a constant at specific T and for a particular solute-solvent pair.



(vi)

Ans AAS is limited to metals

because it relies on the measurement of light absorption by free metal atoms in gas phase. Metals have lower ionization energies, readily form stable atomic species, and more amenable to AAS analysis compared to non-metals. The instrumentation and principles of AAS

are optimized for metal analysis.

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Ans In AAS, chemical interference can occur when elements or compounds in a sample matrix affect the accuracy of the analysis by competing for the same wavelength of light used to measure the target analyte's absorption. Here an example:

During determination of Zn in the presence of Cu by AAS, Cu like zinc can also absorb light at the same wavelength used for zinc analysis. When both Zn and Cu are present in the sample, they will compete for the absorption of light, making it challenging to accurately quantify the zinc concentration. As a result, the AAS analysis may overestimate the zinc content due to the interference from copper.

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(v)

Ans Anion exchange resins have following characteristics:

- 1: Anion exchange resins (AER) contain functional groups that carry a +ve charge. These functional groups attract and exchange anions in the solution.
- 2: Anion exchange resins are selective for anions. They preferentially bind and exchange +ve charged ions like Cl^- , SO_4^{2-} and NO_3^- and others while allowing cations to pass through without significant interaction.
- 3: Anion exchange resins are typically composed of a porous, polymeric matrix. The matrix is usually made of materials like styrene-divinylbenzene.
- 4: Anion exchange resins can be regenerated by eluting the absorbed anion using a regenerant solution e.g. HCl .
- 5: Anion exchange resins are available in various forms, such as gel type resins and macroporous resins.



(vii)

Ans Packing a gel chromatographic column with dry gel is not suitable for several reasons. Dry gel lacks the essential pore structure and hydration properties required for effective size exclusion chromatography. Without proper swelling and hydration, the gel beads cannot provide the desired molecular size-based separation, resulting in poor resolution and inaccurate results. Moreover, dry gel packing can lead to uneven column packing, channeling, and blockages further compromising the chromatographic performance.

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Ans IEC refers to the maximum amount of ions that an ion exchange resin can adsorb or exchange with ions from a solution. It determines how effectively the

resin can remove or exchange ions from a given volume of solution

It is measured in milliequivalent per gram (meq/g) or equivalent per unit weight.

Generally acid-base titration is used to determine IEC.

$$IEC = \frac{V_{NaOH} \times S_{NaOH}}{W_{dry}}$$

where

V = Volume used for titration

S = Strength of titrant

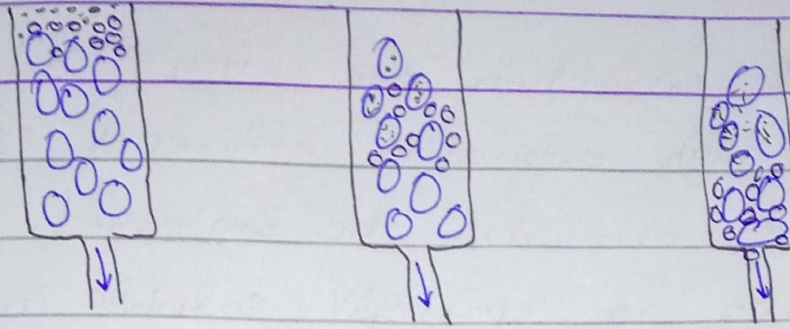
W = Weight of resin (dry)

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(iii)

Ans Gel chromatography, separates molecules primarily based on their size and shape. Smaller molecules enter the pores of matrix and elute later, while larger molecules are excluded and elute earlier, allowing for the determination of relative sizes in a mixture

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- small molecule
- large molecule
- gel particle



(iv)

Ans Simple extraction system:

- In simple extraction system, the extraction process is performed once, where a solvent is used to extract a target compound from a sample or mixture.
- After the extraction, the solvent is typically separated from the extract and the target compound is obtained in the solvent phase.
- Sometimes, complete separation is not achieved.

Multiple extraction system:

- It involves performing the extraction step several times, usually with fresh portions of solvent.
- The goal is to increase the efficiency and completeness of extraction, ensuring that more of the target compound is recovered from the sample.

- Multiple extraction are often used when high extraction efficiency is required or when the ~~target~~ compound is sparingly soluble in the solvent.

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(vi)

Ans The separation efficiency of metal chelates refers to how effectively a chelating agent can extract and separate a specific metal ion from a mixture using a solvent. The efficiency depends on several factors:

- chelating agent
- metal ion
- Solvent selection
- pH & Sol. conditions
- Concentration
- Extraction time

Proper optimization of the factors mentioned above ensures highest separation efficiency and more effective extraction of the target metal ions from the solution.

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Ans SPE is preferred over LLE in many cases due to several advantages, including selectivity, reduced solvent usage, cleaner extracts, automation potential, precision, flexibility with sample sample volumes, ease of handling and compatibility with various analytical separation. These advantages make SPE a more efficient and environmentally friendly extraction method.



(ix)

Ans These are following characteristics of air acetylene flame:

- An air-acetylene flame reaches temperature of 2500 to 2800°C , suitable for atomization and extraction of elements.
- It is reducing flame with excess fuel relative to oxygen, facilitating the reduction of analyte species.
- The flame emits visible light due to thermal excitation and its color can aid in qualitative

element identification.

- It is sensitive for many elements, making it useful for trace elemental analysis.
- Potential spectral interference from sample matrix components require careful correction methods in analytical application.



(xi)

Ans Complexing agents play a vital role in metal extraction processes. They serve four primary functions:

- By forming stable complexes with metal ions, they increase their stability in aqueous phase.
- Their selective binding properties enable the preferential extraction of target metals.
- They prevent undesirable metal precipitation reactions by keeping metal ion in soluble form.
- Complexing agents help control redox conditions, ensuring optimal extraction conditions.

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Ans Solvent extraction is governed by the partition law, which describes the distribution of a solute between two immiscible solvents. The partition coefficient is a key factor, determining the selective extraction of solutes between the two phases. This law allows for multiple extractions and back extraction, facilitating the separation and recovery of specific solutes.



(xiii)

Ans Electrophoresis is one way of analyzing DNA. DNA is arranged in the sequence, for instance, one sequence represents the color of your eyes and another sequence represents the color of your skin. Through electrophoresis, specific DNA sequence can be analyzed, isolated and cloned. The analyzed DNA may be used in forensic investigations.

and paternity tests. Currently, it is used to study the genome of many organisms.



(xv)

Ans The red feathered observed in a nitrous oxide-acetylene flame is caused by the incomplete combustion of acetylene, leading to the formation of carbon soot particles within the flame. These soot particles emit red and orange light, producing the characteristic red feathered appearance.



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